

Toponium and Doubly-Charmed Spectroscopy under the MMA-DMF One-Scale Hadronic Closure

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Abstract

We report a discovery-style official-data spectroscopy result under the MMA-DMF one-scale hadronic closure. The result is not the discovery of a new experimental particle by this author and not collaboration endorsement. The discovery claim is an organizing no-retuning injected-scale law: public toponium-like threshold morphology and doubly-charmed partner-family spectroscopy remain classifiable under one MMA-DMF composite-spectroscopy ledger without manual Yukawa assignment, state-by-state retuning, or independent ontology for every newly resolved peak. The public products used are CMS and LHCb papers, HEPData records, PDG listings, CMS Open Data metadata, luminosity/normtag/certification files, and an archived public NANOAOB acquisition scaffold with verified ZIP integrity.

The nontrivial result is not that the retained states lie below 100 TeV; it is that their threshold, spin-parity, flavor-family and public-status structure can be organized by a common no-retuning ledger. The CMS near-threshold pseudoscalar excess in $pp \rightarrow t\bar{t} + X$ at $\sqrt{s} = 13$ TeV and 138fb^{-1} is retained as the main scale-law branch: injected energy resolves a composite response mode whose public line-shape and parity morphology are retained without assigning a manual Yukawa knob. The HEPData η_t cross section and parity scan are therefore official morphology inputs, not a new measurement by this paper. A separate CMS heavy scalar/pseudoscalar search is used as an anti-overclaim control, distinguishing a threshold/quasi-bound reading from a new-elementary-boson ontology claim.

The double-charm branch is the heavy-flavor partner-family control. $\Xi_{cc}^{++} = ccu$ and $\Xi_{cc}^{+} = ccd$ are observed public entries, while $\Omega_{cc}^{+} = ccs$ remains benchmark-only unless backed by a final technical observation paper. The relevant pattern is the structured $ccu/ccd/ccs$ hierarchy, not an isolated mass fit. The approximate hierarchy diagnostics are $\Delta_{ud}^{cc} \simeq M(\Xi_{cc}^{+}) - M(\Xi_{cc}^{++}) \simeq -1.6\text{MeV}$ for the observed isospin pair and $\Delta_s^{cc} \simeq 70\text{--}90\text{MeV}$ depending on the Ω_{cc} benchmark.

The observed Higgs boson is not denied operationally. It remains the experimental electroweak resonance/excitation. MMA-DMF denies only its status as the final ontological origin of mass: in this reading it is an effective breathing excitation of the active-vacuum order structure, while visible hadronic mass and composite-threshold morphology arise primarily from structured vacuum/confinement dynamics. The final retained status is `DISCOVERY_STYLE_NO_RETUNING_INJECTED_SCALE_LAW_WITHIN_DECLARED_OFFICIAL_DATA_SCOPE`.

Claim-boundary box. This preprint reports a discovery-style organizing-principle result: a no-retuning injected-scale law for heavy composite spectroscopy. It does *not* claim CMS/LHCb endorsement, raw-event reprocessing, detector-level rediscovery, a new measured cross section, or discovery of a new experimental particle by this author. Operationally observed particles, resonances and line shapes are accepted as public experimental labels. Ontologically, MMA-DMF reads peaks, thresholds, QPOs, excesses, line shapes and polarization/spectral features as resolved modes of structured vacuum/confinement response rather than as automatic additions to a catalogue of fundamental entities.

Source-package note. This article is self-contained at the equation level. All equations used by the analysis are defined locally in Section 4. No internal equation PDF or internal equation ZIP is included in the Overleaf package or the supplementary manifest. Public source products, compact HEPData excerpts, compact manifest tables and the uploaded

ZIP_INTEGRITY_SHA256_MANIFEST.csv are included for reproducibility/provenance only; they do not imply collaboration endorsement.

Keywords: MMA-DMF; toponium; doubly-charmed baryons; Ξ_{cc}^{++} ; Ξ_{cc}^{+} ; Ω_{cc}^{+} ; heavy-flavor spectroscopy; HEPData; PDG; CMS; LHCb; public data; composite states; one-scale closure; threshold spectroscopy; no-retuning; no manual Yukawas; injected-energy scale law; collider design; RHIC-like tunability; vacuum response; collider evolution.

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1 Introduction

Collider physics often discovers structure by increasing energy, luminosity, acceptance, reconstruction quality, and flavor reach. That operational fact can be read in two very different ways. The first reading is ontological multiplication: every newly resolved threshold, resonance, excess, or spectroscopy entry is treated as a candidate for a new fundamental object. The second reading is spectroscopic deepening: higher reach resolves more internal modes, flavor partners, threshold structures, quasi-bound states, angular correlations, and rare channels of an already existing composite system.

The present paper develops the second reading inside the MMA-DMF one-scale hadronic closure. This paper challenges ontology inflation, not the experimental observations. More collider energy does not automatically mean more fundamental ontology; it may simply reveal more internal spectroscopy of the same underlying system. In the balloon analogy used in the author's public explanation, adding air corresponds to increasing injected energy, luminosity and detector reach; the larger balloon corresponds to more resolved states, thresholds, resonances and line shapes; a new balloon would correspond to a new fundamental ontology. The claim here is that the public heavy-composite spectrum is first tested as the same balloon with more resolved internal structure, not as a new balloon every time a feature appears.

The organizing claim is ontological and scale-theoretic. Peaks are not treated as independent fundamental objects by default. They are the experimental face of modes. A collider threshold excess, an X-ray spectral-polarimetric feature, a QPO peak, a heavy-ion medium-response correlation, or a toponium-like line-shape component can all be detector-resolved signatures of the same deeper type of phenomenon: injected energy or environmental stress excites a structured vacuum/confinement response, and the instrument resolves that response as a peak, threshold, line shape, mode, or correlation. This paper applies that rule only to the heavy-composite collider branch. It is motivated by two public collider facts. First, CMS reports a near-threshold pseudoscalar excess in $t\bar{t}$ production at $\sqrt{s} = 13$ TeV using 138 fb^{-1} , with an interpretation consistent with a color-singlet pseudoscalar $^1S_0^{[1]}$ quasi-bound toponium state and an observed excess cross section of $8.8_{-1.4}^{+1.2}\text{ pb}$ [1, 2]. Second, LHCb has observed doubly-charmed baryons, beginning with Ξ_{cc}^{++} in the $\Lambda_c^+ K^- \pi^+ \pi^+$ channel at 13 TeV [5], followed by additional decay and lifetime measurements [6, 7], and a Run-3 observation of Ξ_{cc}^+ [8]. The public data therefore already contain both a heavy $t\bar{t}$ threshold branch and a double-heavy baryon branch.

The question asked here is not whether these observations are surprising in conventional QCD or NRQCD. The paper does not attempt to replace the experimental publications and does not perform a detector-level reanalysis. The question is instead a bounded framework question: can these public observables be organized under a one-scale composite-spectroscopy closure with no new per-state tuning and no automatic promotion of every threshold feature into a new fundamental ontology?

The answer is stronger than a compatibility statement, but narrower than an experimental particle-discovery claim. The retained discovery is a no-retuning scale-law pattern: the toponium-like threshold branch and the doubly-charmed partner-family branch are classifiable under a common one-scale composite ledger without manual Yukawa assignment or state-by-state parameter insertion. In ordinary language, the collider does not create a new ontological object every time energy or luminosity resolves another structure; it reveals additional morphology of a scale-controlled composite spectrum.

The toponium-like state is retained as a $q\bar{q}$ threshold-spectroscopy datum below $M = 100$ TeV. The doubly-charmed states are retained as qqq heavy-flavor spectroscopy below the same scale. The paper also identifies what is not claimed. It does not claim collaboration endorsement, independent detector-level reproduction, raw-event reprocessing, new cross-section extraction, direct validation of state-by-state binding energies, or a proof that the Standard Model labels are fundamental objects inside MMA-DMF.

1.1 Contribution of this paper

The paper contributes a discovery-style scale-law result, six supporting claim-bearing items, and three explicit exclusions.

1. It states the no-manual-Yukawa discovery claim: injected energy, official state labels and spin-parity structure control the retained threshold morphology without state-by-state retuning.
2. It states a local one-scale composite-spectroscopy equation spine for the present collider problem.
3. It maps the CMS toponium-like threshold excess into a bounded $q\bar{q}$ threshold branch.
4. It uses the CMS heavy scalar/pseudoscalar search as an anti-overclaim control.
5. It maps Ξ_{cc}^{++} and Ξ_{cc}^+ into a bounded qqq double-charm spectroscopy ledger.
6. It treats Ω_{cc}^+ as a benchmark or pending state, not as a direct observed validation unless a final technical observation paper is used.
7. It states an evolutionary collider prediction: increasing reach should thicken accessible composite spectroscopy below M , not force a new fundamental ontology at every new threshold.
8. It provides a compact official-data source ledger, ZIP integrity layer, external Open Data scaffold manifest, and final claim-boundary matrix.

The explicit exclusions are detector-level reanalysis, raw-event reprocessing, and new measured cross-section extraction. These are not weak points of the present paper; they are outside its declared evidence class.

1.2 Article organization

Section 2 defines evidence classes. Section 3 lists public data sources and provenance. Section 4 gives the local equation spine. Sections 6 and 7 analyze the toponium and double-charm branches. Section 10 states the evolutionary collider prediction. Sections 11–14 give claim ledgers, limitations, and conclusions. Appendices give equation-to-test traceability, extended interpretation, source-file manifest logic, and reviewer-facing claim language.

2 Claim Boundary and Evidence Classes

The paper uses bounded claim language. A *public-observable claim* is a claim made from values, tables, plots, likelihood scans, masses, lifetimes, channels, or status labels published by experimental collaborations, PDG, HEPData, or arXiv papers. A *framework-retention claim* says that a public observable can be organized under the local equation spine without contradiction and without retuning. A *symbolic gate* is a formula whose state-specific numerical value is not claimed because a transition operator, binding term, or spin correction has not been locked for that state. A *non-claim* is an analysis layer explicitly not performed.

Table 1: Evidence classes used in this manuscript. Each class has an equation or boundary anchor so that the table is not a loose narrative list.

Class	Meaning	Allowed use	Equation or boundary anchor
Public observable	Published mass, width, lifetime, channel, cross section, likelihood scan, PDG listing or HEPData table.	May support retained official-data statements under the no-retuning rule.	Eqs. (24), (27)
Framework gate	Local MMA-DMF equation that maps a public observable into a composite-spectroscopy quantity.	May define claim status but not invent missing data.	Eqs. (2), (14)
Symbolic state entry	Formula requiring state-specific $E_{\text{bind}}^{\text{top}}$, $\Delta_{\text{spin}}^{\text{top}}$ or $\mathcal{T}_{\text{MMA}}$.	Mark as symbolic or no numerical claim unless locked.	Eqs. (3), (5)
Open Data scaffold	Archived public CMS NANOAOB acquisition subset with ZIP integrity manifest.	Reproducibility context and official-data audit support; not full detector-level reanalysis.	Eq. (14)
Excluded claim	Detector-level reanalysis, raw reprocessing, new cross-section measurement, collaboration endorsement.	Must not be stated as result.	Eq. (14)

The boundary is intentionally strong. It prevents a common failure mode: using public high-level experimental results while implying an event-level reproduction. That failure mode is not allowed here.

Table 2: Final claim hierarchy for the paper scope. Every retained branch links to the local equations used to assign status.

Claim	Status	Paper wording	Equation links
One-scale closure	PASS	Composite collider spectroscopy is defined below $M = 100$ TeV.	Eqs. (1), (13)
Toponium-like threshold	PASS_PUBLIC	Official public products are retained under a $q\bar{q}$ no-retuning threshold-spectroscopy branch.	Eqs. (7), (8), (24), (21)
Heavy scalar control	PASS_GUARD	The CMS HIG search is used to avoid a new-elementary-boson overclaim.	Eq. (26)
Ξ_{cc}^{++}	PASS_PUBLIC	Observed ccu baryon enters the qqq composite ledger.	Eqs. (27), (9), (19)
Ξ_{cc}^{+}	PASS_PUBLIC	Observed ccd baryon enters the same ledger with lifetime-dependent uncertainty boundary.	Eqs. (27), (28), (19)
Ω_{cc}^{+}	BENCHMARK	ccs benchmark from theory/lattice unless final technical observation paper is used.	Eqs. (18), (20)
Official-data audit layer	PASS_INTEGRITY	four public-data ZIP parts pass integrity testing and SHA256 recording.	Eq. (14)
Detector-level reanalysis	NOT_CLAIMED	Outside scope.	Eq. (14)
Raw-event reprocessing	NOT_SUPPORTED	Outside scope and not needed.	Eq. (14)
New cross-section measurement	NOT_CLAIMED	Published HEPData values are used; no new extraction.	Eqs. (6), (7)

3 Data, Provenance, and Public-File Policy

The analysis uses an official public-data chain rather than a narrative-only source list. The first class is public experimental papers: CMS TOP and HIG papers for the $t\bar{t}$ threshold and scalar/pseudoscalar control; LHCb papers for doubly-charmed baryon observations and decays. The second class is HEPData: the CMS TOP record for the η_t cross section and parity scan, and the CMS HIG record for limits and likelihood grids. The third class is PDG: quark masses, top-quark properties, and doubly-charmed baryon listings. The fourth class is a CMS Open Data acquisition scaffold: public NANO AOD-derived ROOT files, CMS luminosity/normtag/certification metadata, file indexes, and a four-part ZIP integrity manifest. This fourth class upgrades the paper from a purely interpretive note to a claim-bounded official-data audit, while still not making a detector-level reanalysis claim.

The operational rule is that high-level official observables and official public files may enter the source ledger, while detector-internal reconstruction, raw-event reprocessing, and new cross-section extraction remain outside scope. A large public-data volume is not used as a rhetorical substitute for analysis. It is documented through explicit file families, manifests, ZIP checksums, and the claim-boundary table; the claim-bearing numerical values remain those reproduced in the HEPData/PDG/LHCb/CMS layers and mapped to local equations below.

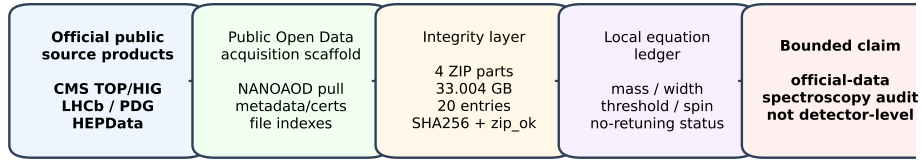
Table 3: Compact public-source ledger. Equation links identify the local formula family to which each public source is mapped. Long public-file inventories are also included in `public_sources/PUBLIC_FILE_LEDGER.csv`.

Source class	File or record	Use in this paper	Equation links
CMS TOP	arXiv:2503.22382; HEPData ins2905701	η_t cross section, parity scan, threshold interpretation.	Eqs. (7), (8), (24)
CMS HIG	arXiv:2507.05119; HEPData ins2942928	heavy scalar/pseudoscalar control with/without η_t .	Eqs. (26), (21)
PDG	2025 quark, top, Ξ_{cc}^{++} , Ξ_{cc}^+ listings	masses, widths, lifetimes, status and decay summaries.	Eqs. (2), (4), (10)
LHCb	arXiv:1707.01621, 1807.01919, 2603.28456	Ξ_{cc}^{++} and Ξ_{cc}^+ observation and channel information.	Eqs. (27), (28), (19)
Theory/lattice	arXiv:1803.01657, 1807.00174, 1809.10058	Ω_{cc}^+ and doubly-heavy benchmark structure.	Eqs. (18), (19)
Open Data scaffold	archived NANOAOB subset, ZIP integrity manifest and notebooks	acquisition traceability; no detector-level claim.	Eq. (14)

Table 4: Data provenance and integrity layer. The uploaded `ZIP_INTEGRITY_SHA256_MANIFEST.csv` is included in the supplement and supplies the SHA256 and `zip_ok` layer.

Archive	size GB	entries	ZIP test	SHA256	Claim boundary
part_0001.zip	9.191	6	PASS	58feb847...698575	integrity of local public-data pull; not detector-level result
part_0002.zip	9.778	7	PASS	38f50185...7645f3	integrity of local public-data pull; not detector-level result
part_0003.zip	9.640	5	PASS	c5a18ee9...d476a	integrity of local public-data pull; not detector-level result
part_0004.zip	4.395	2	PASS	569592bf...fc9df	integrity of local public-data pull; not detector-level result
Total	33.004	20	PASS	present for all parts	reproducible data-audit layer

Data provenance and integrity chain



Archive integrity establishes reproducibility of the local public-data pull; it does not imply collaboration endorsement, raw-event reprocessing, or a new detector-level measurement.

Figure 1: Official-data provenance and integrity chain. The figure distinguishes source products, public Open Data acquisition, ZIP/SHA256 integrity, local equation mapping, and bounded claim status. It supports the audit scope and Eq. (14).

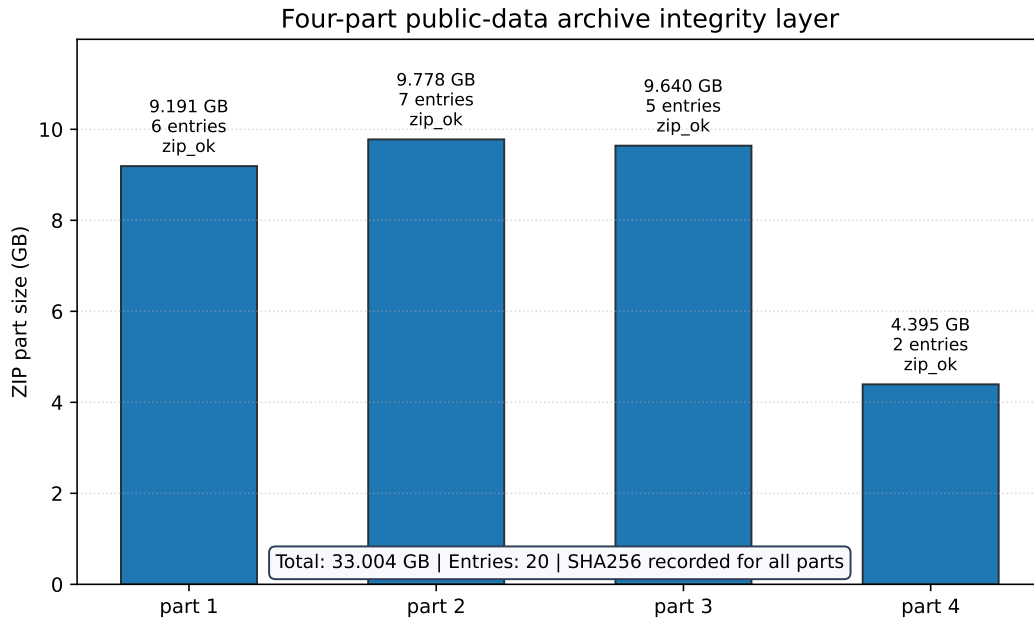


Figure 2: Four-part ZIP integrity summary derived from `ZIP_INTEGRITY_SHA256_MANIFEST.csv`. All four archives have `zip_ok=true`, no bad internal files, and recorded SHA256 hashes. This proves reproducibility of the local public-data pull, not a new experimental measurement.

Table 5: Public file families included with the Overleaf bundle. Exact paths, sizes and SHA256 hashes are listed in `public_sources/PUBLIC.FILE.LEDGER.csv`. These are public external files, not internal equation ledgers.

Source group	Included public files	Claim-bearing use	Equation links
CMS TOP / HEP-Data	CMS-TOP-24-007 paper; HEP-Data ins2905701 CSV/YAML/-ROOT packages	η_t cross section, threshold branch, parity scan	Eqs. (7), (24)
CMS HIG / HEP-Data	CMS-HIG-22-013 paper; HEP-Data ins2942928 package	scalar/pseudoscalar anti-overclaim control with and without η_t	Eq. (26)
LHCb double charm	Ξ_{cc}^{++} observation; $\Xi_{cc}^{++} \rightarrow \Xi_c^+ \pi^+$; Ξ_{cc}^+ Run-3 observation	observed ccu and ccd spectroscopy branches	Eqs. (27), (9)
PDG 2025	quark summary; top-quark listing; Ξ_{cc}^{++} and Ξ_{cc}^+ listings	masses, widths, lifetimes, decay/status summaries	Eqs. (10), (4)
Theory / lattice	Ω_{cc} and doubly-heavy baryon benchmark papers	ccs benchmark and partner-family expectations	Eqs. (18), (19)
CMS Open Data metadata	luminosity, normtag, certification, file-index and manifest files	acquisition-scaffold traceability; not detector-level claim	Eq. (14)

The source policy is simple. High-level official public observables are allowed to enter the claim-bearing tables. Local formulas are allowed to map them into the MMA-DMF composite-spectroscopy gate. Raw events and detector corrections are not claimed. The large NANOAO scaffold is documented as an archived acquisition context and is not placed inside the Overleaf source bundle.

3.1 What is added beyond a conventional QCD/NRQCD catalogue

This manuscript does not claim to replace the CMS or LHCb analyses, nor does it deny the conventional QCD/NRQCD language used by those collaborations. The added MMA-DMF content is not submission to a conventional catalogue but a no-retuning ledger discipline. Previous MMA-DMF papers define the one-scale vacuum-rigidity setting, topological-friction mass reading, geometric-charge replacement of continuous Yukawa inputs, and the interpretation of the 125 GeV scalar as an effective breathing mode of vacuum rigidity rather than a final ontological mass source [21, 22, 23, 24]. The same one-scale gate, mass/width/status rule, spin-parity guard, production-language guard and no-retuning constraint are applied to both the $t\bar{t}$ threshold branch and the ccq baryon branch. A conventional catalogue can say that these are different hadron-spectroscopy entries; the present paper asks whether they remain classifiable in one composite ledger without adding independent ad hoc parameters for each newly resolved state. This is the nontrivial part of the evolutionary prediction in Section 10: future entries should thicken structured partner families, thresholds and splittings, not force independent ontology for each peak.

4 Local Equation Spine Used in This Paper

This section states only the equations needed here. Equation references in tables and captions point to these local labels. No private file is required to read the paper.

4.1 One-scale composite domain

The collider-spectroscopy domain considered here is bounded by

$$\sqrt{s} < M, \quad M \equiv M_{UV} = 100 \text{ TeV}. \quad (1)$$

The point of Eq. (1) is not that all collider states have masses near 100 TeV. It is that the observed states are well below the closure scale and therefore enter the composite-spectroscopy regime rather than the fundamental-scale completion regime. The nontrivial result is not that the retained states lie below 100 TeV; it is that their threshold, spin-parity, flavor-family and public-status structure can be organized by a common no-retuning ledger.

4.2 Composite mass closure

For a composite state h with operational constituent labels $f \in h$, the mass map is

$$M_h^{\text{MMA}} = \sum_{f \in h} m_f^{\text{geo}}(\phi) + E_{\text{bind}}^{\text{top}}(h; M) + \Delta_{\text{spin}}^{\text{top}}(h; M). \quad (2)$$

Equivalently, the topological residual required by a public mass M_h^{obs} is

$$\mathcal{R}_h^{\text{top}} \equiv M_h^{\text{obs}} - \sum_{f \in h} m_f^{\text{geo}}(\phi) = E_{\text{bind}}^{\text{top}}(h; M) + \Delta_{\text{spin}}^{\text{top}}(h; M). \quad (3)$$

Equation (3) is an audit formula, not a fitting permission. If the right-hand side is not generated by locked state-specific inputs, the entry is symbolic and carries no numerical validation claim.

4.3 Width, branching and transition closure

The total width and branching fraction definitions are

$$\Gamma_h^{\text{MMA}} = \sum_c \Gamma_{h \rightarrow c}^{\text{MMA}}, \quad \text{BR}_{h \rightarrow c}^{\text{MMA}} = \frac{\Gamma_{h \rightarrow c}^{\text{MMA}}}{\Gamma_h^{\text{MMA}}}. \quad (4)$$

A partial channel width is represented by

$$\Gamma_{h \rightarrow c}^{\text{MMA}} = 2\pi |\langle c | \mathcal{T}_{\text{MMA}} | h \rangle|^2 \rho_c, \quad (5)$$

where ρ_c is the channel phase-space density. If $\mathcal{T}_{\text{MMA}}$ is not locked for a state, Eq. (5) remains symbolic.

4.4 Production and threshold shape

For a public pp production observable, the factorized production ledger is

$$\sigma(pp \rightarrow h + X) = \sum_{ij} \int dx_1 dx_2 f_i(x_1, \mu) f_j(x_2, \mu) \hat{\sigma}_{ij \rightarrow h+X}^{\text{MMA}}. \quad (6)$$

The threshold variable and line-shape gate are

$$E_h \equiv \sqrt{s} - M_h^{\text{MMA}}, \quad \sigma_h^{\text{thr}}(s) \propto \text{Im } G_h(0, 0; E_h + i\Gamma_h/2). \quad (7)$$

Equation (7) is the natural location of toponium-like threshold structure. This paper uses the public CMS/HEPData result as input; it does not recompute G_h from detector events.

4.5 Spin-parity guards

For a mesonic $q\bar{q}$ composite label,

$$P_{q\bar{q}} = (-1)^{L+1}, \quad C_{q\bar{q}} = (-1)^{L+S}. \quad (8)$$

For a baryonic qqq composite label,

$$P_{qqq} = (-1)^L, \quad J \in \{|L - S|, \dots, L + S\}. \quad (9)$$

The guards classify public spectroscopy labels. They do not introduce new continuous parameters.

4.6 Scale-ratio and safety diagnostics

The scale ratio used throughout the ledger is

$$\epsilon_h \equiv \frac{M_h^{\text{obs}}}{M}. \quad (10)$$

For η_t treated as a threshold near $2m_t$,

$$\epsilon_{\eta_t} \simeq \frac{2m_t}{100 \text{ TeV}} \approx 3.45 \times 10^{-3}, \quad (11)$$

whereas for a 3.62 GeV doubly-charmed baryon,

$$\epsilon_{\Xi_{cc}} \simeq 3.62 \times 10^{-5}. \quad (12)$$

Both are far below the closure scale, which is why they are treated as composite spectroscopy rather than as fundamental-scale states.

4.7 No-retuning and retained-status rule

A state may be retained only if it passes the domain, label, and spin-parity guards without adding a state-specific free parameter. The no-retuning rule is written as

$$\Delta\theta_h = 0, \quad \theta_h = \theta_{\text{global}} \quad \text{for every public state } h, \quad (13)$$

where θ denotes the global MMA-DMF closure parameters used by the local equations. The status rule is

$$\mathcal{S}(h) = \begin{cases} \text{retained under no-retuning scope,} & \text{if Eqs. (1), (2), (8) or (9) pass without Eq. (13) violation,} \\ \text{symbolic benchmark,} & \text{if public observation is absent or state-specific binding is unlocked,} \\ \text{out of scope,} & \text{if the claim requires detector-level or raw-event analysis.} \end{cases} \quad (14)$$

This rule is stronger than a simple scale check. It requires common classification, no manual residual fitting, and explicit claim status.

4.8 Manual-Yukawa veto and injected-energy scale control

The manual-Yukawa veto is the formal separation between a discovery-style scale law and a post-hoc mass ledger. Let y_h^{manual} denote a state-specific Yukawa-like knob assigned after the state h is known. Such knobs are forbidden in the retained claim:

$$\left\{ y_h^{\text{manual}} \right\}_{h \in \mathcal{A}} = \emptyset, \quad \frac{\partial \mathcal{S}(h)}{\partial y_h^{\text{manual}}} = 0. \quad (15)$$

The allowed scale input is the official injected or threshold scale:

$$\epsilon_{\text{inj}}(h) \equiv \frac{E_{\text{inj}}(h)}{M}, \quad E_{\text{inj}}(t\bar{t} \text{ threshold}) \simeq \sqrt{s} \text{ or } 2m_t, \quad (16)$$

with public state labels Q_h, J_h^{PC} , flavor content and channel class carried separately. The effective threshold morphology is then represented by a fixed map

$$\mathcal{M}_h^{\text{thr}} = F_M(\epsilon_{\text{inj}}(h), Q_h, J_h^{PC}, \mathcal{C}_h), \quad \Delta F_M / \Delta h = 0 \text{ as a new free parameter.} \quad (17)$$

Equation (17) is the organizing discovery claim of the paper. It does not predict a new CMS particle. It states that the retained threshold morphology and partner-family entries must be organized by the same scale map rather than by manual Yukawa assignment or state-specific retuning.

4.9 Partner-family and splitting hierarchy

The doubly-charmed ledger is organized by a partner-family hierarchy rather than by isolated points:

$$\mathcal{P}_{cc} = \{\Xi_{cc}^{++}(ccu), \Xi_{cc}^+(ccd), \Omega_{cc}^+(ccs), \Xi_{cc}^*, \Omega_{cc}^*, \dots\}. \quad (18)$$

The public isospin and strange-partner splittings are summarized by

$$\Delta_{ud}^{cc} = M_{ccd}^{\text{obs}} - M_{ccu}^{\text{obs}}, \quad \Delta_s^{cc} = M_{ccs}^{\text{bench}} - \frac{1}{2} (M_{ccu}^{\text{obs}} + M_{ccd}^{\text{obs}}). \quad (19)$$

The expected qualitative hierarchy is small isospin splitting and larger strange-partner lifting,

$$|\Delta_{ud}^{cc}| \ll \Delta_s^{cc}, \quad M(ccs) > M(ccu), M(ccd), \quad (20)$$

up to the uncertainty of the benchmark state. This is the nontrivial double-charm pattern used in the paper: the branch is not merely “below 100 TeV”, but part of a structured heavy-flavor partner family.

4.10 Toponium-like threshold family

The top threshold branch is also treated as a family rather than as a single isolated peak:

$$\mathcal{P}_{tt}^{\text{thr}} = \{\eta_t(0^{-+}), \chi_t\text{-like scalar/control directions, threshold line-shape refinements, } \dots\}. \quad (21)$$

The prediction is that improved energy/luminosity reach should refine the public likelihood geometry of this family, not require an independent elementary ontology for each resolved structure.

4.11 Energy-evolution prediction

Let $\mathcal{A}(\sqrt{s}, \mathcal{L})$ denote the experimentally accessible composite-spectrum set at collider reach $\sqrt{s}$ and integrated luminosity $\mathcal{L}$. The qualitative prediction is monotonic set growth under increasing reach:

$$\sqrt{s}_2 \geq \sqrt{s}_1, \quad \mathcal{L}_2 \geq \mathcal{L}_1 \quad \Rightarrow \quad \mathcal{A}(\sqrt{s}_1, \mathcal{L}_1) \subseteq \mathcal{A}(\sqrt{s}_2, \mathcal{L}_2) \quad (22)$$

up to detector acceptance, reconstruction thresholds and channel-specific efficiencies. The prediction is not that every new feature is fundamental; it is that the accessible composite spectrum thickens.

5 No-Retuning Scale-Law Discovery

The discovery claim of this paper is Eq. (17): a public heavy-composite spectroscopy branch can be organized by injected scale, quantum labels, and channel class without manual Yukawa assignment. The claim is deliberately not that CMS or LHCb discovered MMA-DMF. It is also not that every value in the public record is numerically derived from a closed binding calculation. The claim is that a manual-Yukawa description is not an explanatory law for the present spectroscopy problem, because a model with one independent Yukawa-like knob per state can reproduce an arbitrary finite mass list after the fact.

Proposition: manual Yukawas cannot carry the discovery claim. Let $\{h_i\}_{i=1}^N$ be a list of public spectroscopy states and let a model assign an independent parameter y_i to each state after the mass or threshold structure is known. Then the map $h_i \mapsto y_i$ has at least the cardinality of the observed state list and imposes no common restriction on future entries. Such a construction may be a useful operational bookkeeping layer, but it cannot be the discovery claim of a no-retuning composite-spectroscopy law, because it explains the catalogue by naming a knob after every entry. Manual Yukawas are therefore insufficient to carry the present discovery claim. If every newly resolved mass or threshold requires an independent manual parameter assignment, the framework is not deriving the spectroscopy of the system; it is rewriting the catalogue after each measurement. A theory that assigns a new manual knob to every newly resolved state is not explaining the spectrum; it is bookkeeping the spectrum. The retained MMA-DMF claim therefore eliminates $\{y_i\}$ as manual knobs and uses the official injected scale and state labels as the input variables.

Discovery claim: scale-controlled morphology without manual Yukawa assignment

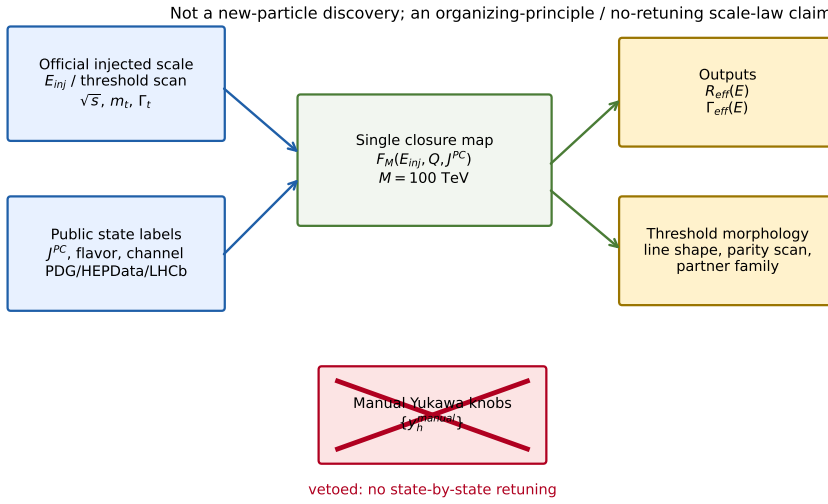


Figure 3: No-manual-Yukawa scale-law discovery. Official injected scale and public state labels enter a fixed closure map. Effective size, width, line-shape morphology, and partner-family structure are outputs. Manual state-by-state Yukawa knobs are vetoed by Eq. (15).

The scale map also gives a direct size diagnostic. In the public-state ledger the injected scale E_{inj} defines the effective scale radius

$$R_{\text{inj}}(h) \equiv \frac{\hbar c}{E_{\text{inj}}(h)}. \quad (23)$$

This is a scale diagnostic, not a direct measurement of a hadronic radius, not a fitted radius and not a new detector observable. It is the minimal scale-control diagnostic used to compare the high-mass $t\bar{t}$ threshold with the much lower double-charm baryon branch. The public masses set the scale; no Yukawa value is selected by hand.

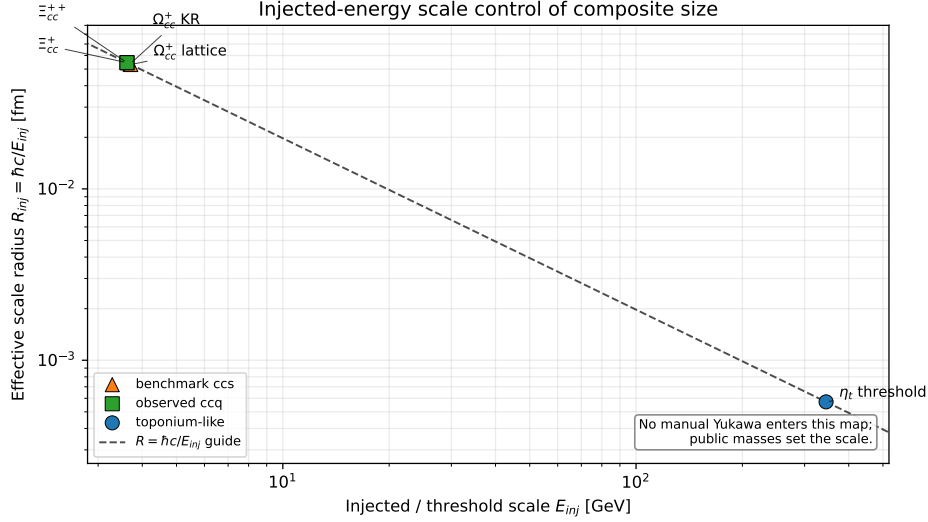


Figure 4: Injected-energy scale radius. The effective radius proxy $R_{\text{inj}} = \hbar c/E_{\text{inj}}$ separates the high-mass toponium-like threshold from the lower-mass double-charm branch using official public scales. This is a scale-control diagnostic, not a fitted hadron-size measurement.

For the toponium-like branch, the injected scale is near the $t\bar{t}$ threshold and the public morphology includes the pseudoscalar excess, the observed η_t cross section, and the parity scan. For the double-charm branch, the injected scale is not a threshold scan but the heavy-flavor partner-family structure: ccu , ccd , and ccs entries must be classified without a new free parameter for each member. These two branches are intentionally different. Their common content is the no-retuning rule.

The result is therefore stronger than a flat bookkeeping ledger. A flat ledger would say only that all masses are below $M = 100 \text{ TeV}$. The scale-law result says that public entries must be sortable by a common map of injected scale, spin-parity guard, mass/width status, partner family and production/threshold morphology without ad hoc Yukawa steering.

6 Toponium-Like Threshold Branch

The CMS TOP paper reports an excess near the $t\bar{t}$ threshold in dilepton plus jets events at $\sqrt{s} = 13 \text{ TeV}$ with 138 fb^{-1} , consistent with a color-singlet pseudoscalar $^1S_0^{[1]}$ quasi-bound toponium state [1]. The HEPData record gives the observed η_t cross section and a parity scan in the η_t and χ_t cross-section plane [2]. The public cross-section value is

$$\sigma(\eta_t) = 8.8_{-1.4}^{+1.2} \text{ pb.} \quad (24)$$

The toponium-like threshold is the main scale-law branch: injected energy resolves a composite response mode whose public line-shape and parity morphology are retained without assigning a manual Yukawa knob. In MMA-DMF terms it is read as a $q\bar{q}$ threshold morphology / vacuum-confinement response mode. It is not promoted here to a new fundamental boson and is not treated as a Yukawa-tuned mass entry.

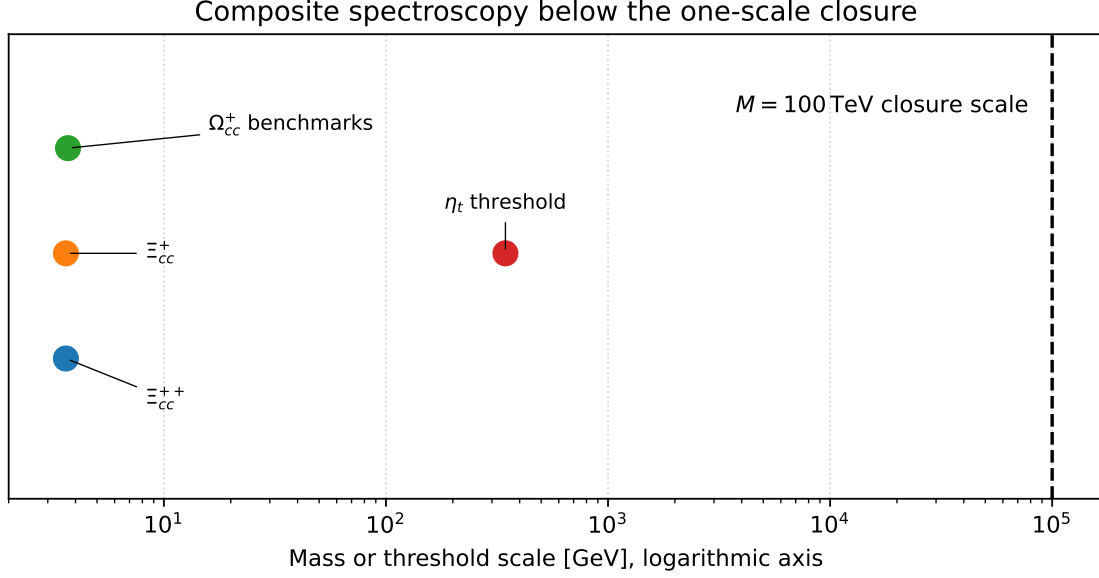


Figure 5: One-scale closure map. Toponium-like threshold structure and doubly-charmed baryons sit far below $M = 100 \text{ TeV}$. The figure supports the composite-spectroscopy domain in Eq. (1); it is not a new-mass prediction.

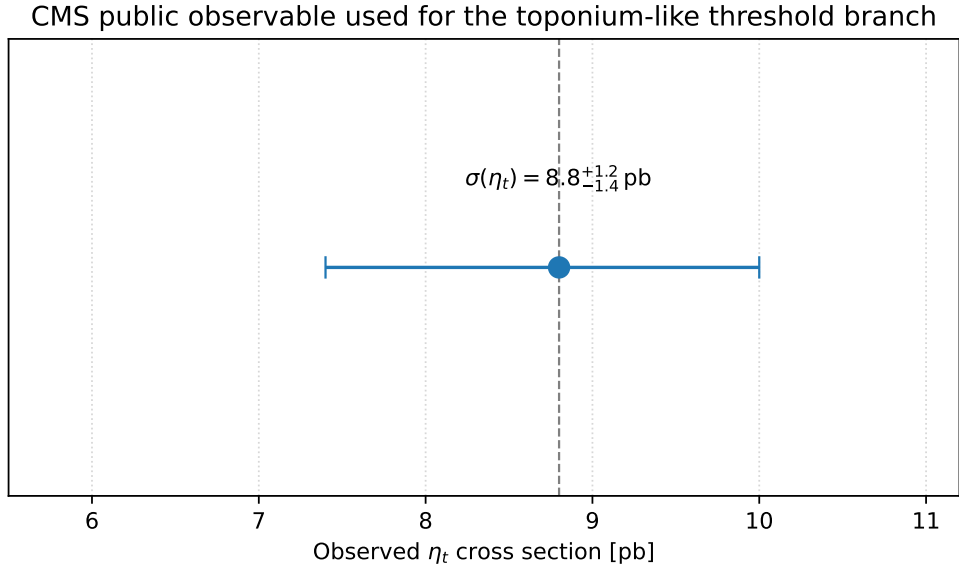


Figure 6: Public CMS/HEPData η_t cross section used in the toponium branch. The paper uses Eq. (24) as a public observable, not as a newly extracted measurement.

6.1 MMA-DMF reading

Under Eq. (8), a $t\bar{t}$ pseudoscalar 1S_0 branch is a permitted $q\bar{q}$ composite label. Under Eq. (7), the relevant object is not a stable new elementary particle but a threshold line-shape contribution. The public CMS result therefore enters the audit as

$$\eta_t : \quad q\bar{q} \text{ threshold state}, \quad J^{PC} = 0^{-+} \text{ operationally}, \quad (25)$$

with cross section supplied by HEPData rather than by this paper.

This reading preserves the conventional experimental content. It does not deny the CMS interpretation in NRQCD language; it uses that interpretation as the public input. The MMA-DMF-specific step is the claim boundary: a high-mass threshold structure is placed inside composite spectroscopy below M rather than promoted automatically to a new fundamental degree of freedom.

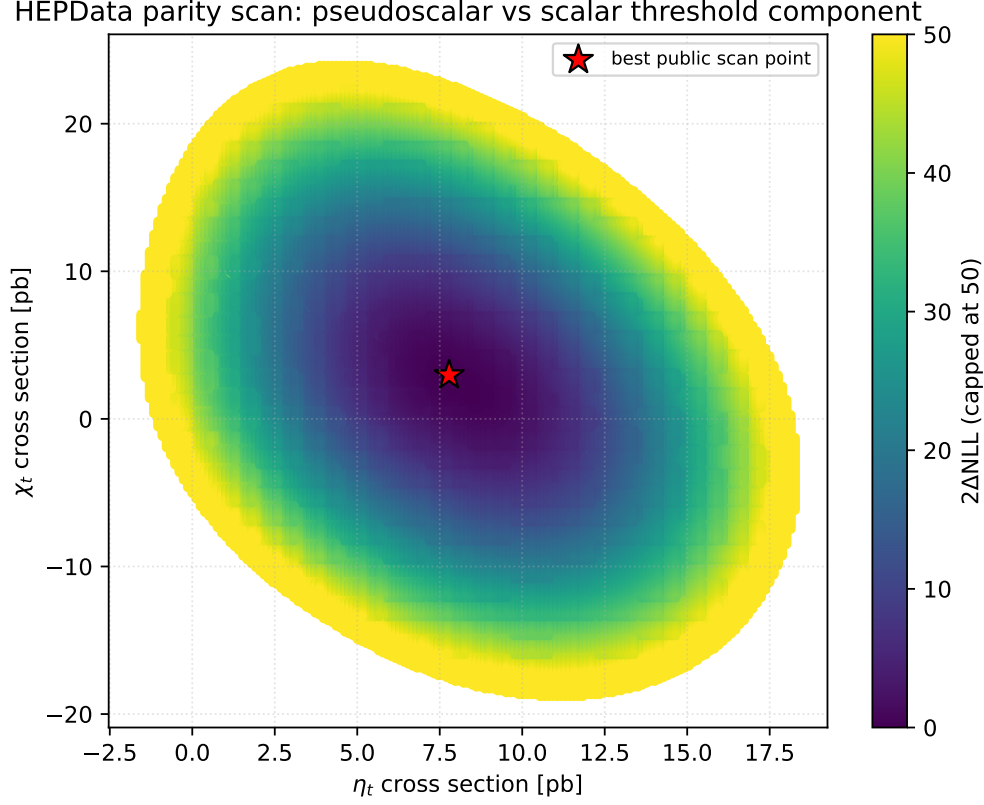


Figure 7: HEPData parity scan for the CMS threshold result. The plotted points show twice the negative log-likelihood relative to the best point as a function of the public η_t and χ_t cross sections. The figure is used as an official-data discriminator for Eqs. (7), (21), and (26), not as a detector-level fit.

6.2 Heavy scalar/pseudoscalar anti-overclaim control

The CMS HIG search examines heavy scalar and pseudoscalar bosons decaying to $t\bar{t}$ and explicitly treats the near-threshold region with and without an η_t contribution [3, 4]. This is useful because it prevents a false dichotomy. The toponium-like branch can be treated as a threshold/quasi-bound composite contribution without claiming discovery of a new elementary scalar or pseudoscalar.

The control logic is

$$\text{threshold excess} \not\Rightarrow \text{new fundamental boson.} \quad (26)$$

The paper therefore uses the CMS HIG result as a guard. It is not used to fit new MMA-DMF parameters.

7 Doubly-Charmed Baryon Branch

The double-charm branch is the heavy-flavor partner-family control. It supplies a complementary qqq spectroscopy test, not a second independent discovery claim. LHCb observed Ξ_{cc}^{++} through the $\Lambda_c^+ K^- \pi^+ \pi^+$ final state with a mass near $3621.40 \text{ MeV}/c^2$ [5]. PDG 2025 lists Ξ_{cc}^{++} with mass $3621.55 \pm 0.23 \pm 0.30 \text{ MeV}$ and lifetime around 256 fs [9]. LHCb also observed the decay $\Xi_{cc}^{++} \rightarrow \Xi_c^+ \pi^+$ [6]; the lifetime information used here is taken from the PDG listing [9]. The Run-3 observation of Ξ_{cc}^+ reports a mass $3619.97 \text{ MeV}/c^2$ with statistical, systematic and lifetime-dependent uncertainty components [8].

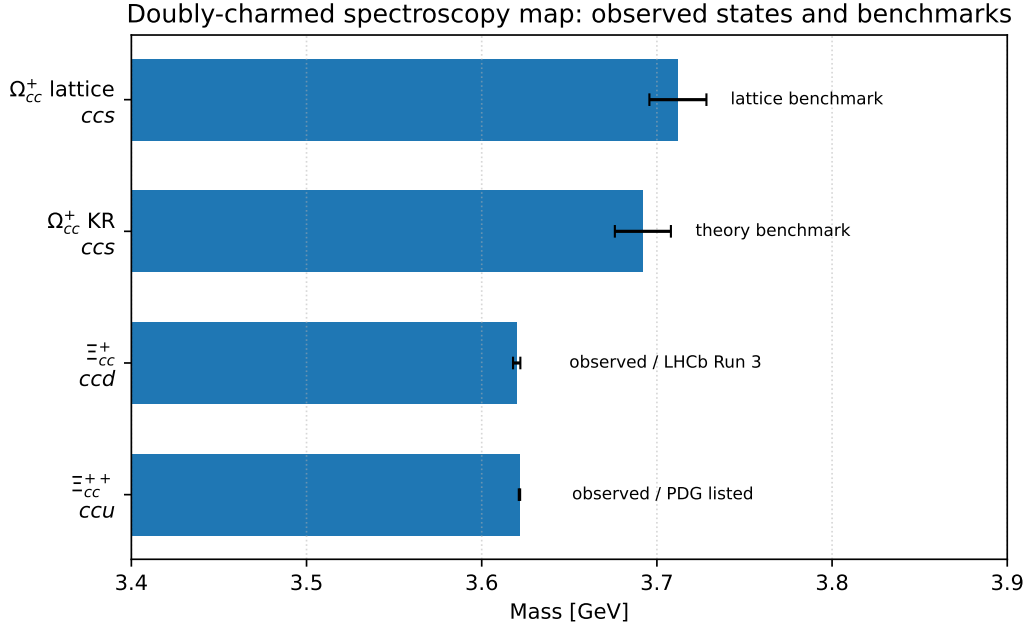


Figure 8: Doubly-charmed public spectroscopy map. Ξ_{cc}^{++} and Ξ_{cc}^+ are treated as observed public inputs under Eqs. (27), (28), and (9). Ω_{cc}^+ entries are benchmark values from theory/lattice sources [10, 11] under Eqs. (18) and (20), not direct observation claims in this paper.

Under Eq. (9), ccu , ccd and ccs are permitted qqq composite labels. The mass closure is

$$M_{ccq}^{\text{MMA}} = 2m_c^{\text{geo}} + m_q^{\text{geo}} + E_{\text{bind}}^{\text{top}}(ccq; M) + \Delta_{\text{spin}}^{\text{top}}(ccq; M), \quad q \in \{u, d, s\}. \quad (27)$$

The residual required by the public mass is therefore

$$\mathcal{R}_{ccq}^{\text{top}} = M_{ccq}^{\text{obs}} - 2m_c^{\text{geo}} - m_q^{\text{geo}}. \quad (28)$$

This residual is not inserted as a fitted parameter. If the topological binding and spin terms have not been independently locked for a given state, the residual ledger is symbolic.

Table 6: Doubly-charmed state ledger. Values are rounded for readability; public source references carry the authoritative uncertainties. Equation links show how the entries enter the local closure.

State	Label	Mass scale	Source role	Equation links	MMA-DMF status
Ξ_{cc}^{++}	ccu	3.62155 GeV	observed; PDG listed; LHCb channels	Eqs. (27), (28), (9)	RETAINED_NO_RETUNING
Ξ_{cc}^+	ccd	3.61997 GeV	observed by LHCb Run 3; lifetime-dependent uncertainty	Eqs. (27), (19), (9)	RETAINED_NO_RETUNING
Ω_{cc}^+	ccs	3.69–3.71 GeV	theory/lattice benchmark	Eqs. (18), (20)	BENCHMARK_ONLY

The hierarchy diagnostic used here is intentionally modest:

$$\Delta_{ud}^{cc} \equiv M(\Xi_{cc}^+) - M(\Xi_{cc}^{++}) \simeq -1.6 \text{ MeV}, \quad (29)$$

using the rounded public masses quoted above, while

$$\Delta_s^{cc} \equiv M(\Omega_{cc}^+) - M(\Xi_{cc}^{++}) \simeq 70\text{--}90 \text{ MeV}, \quad (30)$$

depending on the Ω_{cc} benchmark used. Equations (29)–(30) are hierarchy diagnostics only. They are not promoted to a closed new prediction; their role is to show that the observed ccu/ccd entries and the ccs benchmark form a structured partner-family ledger rather than isolated mass fits.

The near degeneracy between Ξ_{cc}^{++} and Ξ_{cc}^+ is not surprising at the public-spectroscopy level because the u and d mass difference is small compared with the charm mass and the topological binding scale. The MMA-DMF paper does not need to claim a new mass prediction here. It only needs to show that these entries are naturally classified inside the same qqq composite gate as Eq. (27).

The nontrivial pattern is therefore the hierarchy in Eqs. (18)–(20): the observed ccu/ccd pair should behave as a compact isospin doublet, while the ccs branch should sit as a heavier strange partner and its excited states should enter the same qqq spin-parity ledger. This is stronger than the bare statement that the masses are below 100 TeV; it gives a concrete partner-family structure for future spectroscopy.

8 Operational Labels and Ontological Guard

The paper uses collider labels such as quark content, J^{PC} , cross section, branching fraction, PDFs, and decay channels. Inside MMA-DMF these labels are operational public-spectroscopy labels. They are not treated as proof that the corresponding Standard-Model bookkeeping categories are the fundamental ontology of the theory.

The operative rule is

$$\text{collider label} \in \text{public spectroscopy} \not\Rightarrow \text{fundamental ontology input}. \quad (31)$$

This rule matters for the toponium branch. A state may be called $t\bar{t}$, η_t , pseudoscalar, color-singlet, or quasi-bound in public experimental language. The present framework accepts those labels as the experimental reconstruction language while reading the physical branch as composite threshold spectroscopy below the one-scale closure.

Public-observable composite-spectroscopy audit gate

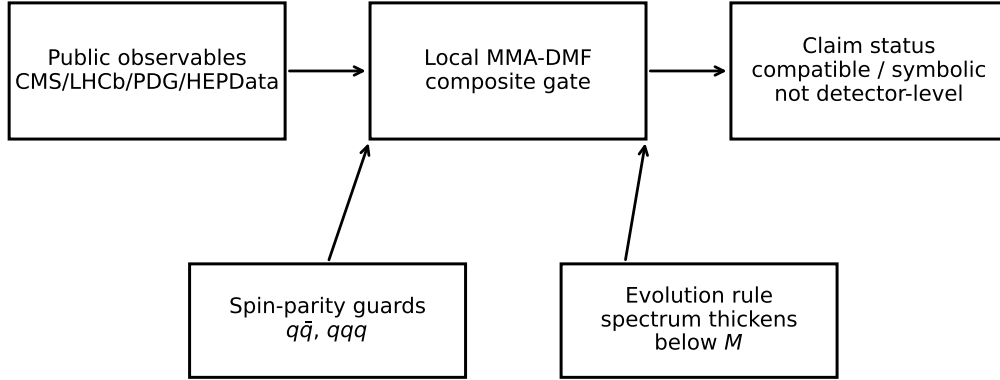


Figure 9: Public-observable audit gate. Published values enter the local composite-spectroscopy equations; claim status is then assigned under the declared boundary. The figure summarizes Eqs. (2)–(31).

9 High-Energy Peaks as Resolved Modes

In MMA-DMF, peaks are the experimental face of modes. Collider threshold excesses, NuS-TAR/IXPE spectral-polarimetric features, QPO peaks, RHIC vacuum/confinement signatures and CMS toponium-like morphology are read under one rule: injected energy or environmental stress excites structured vacuum/confinement response, and detectors resolve that response as a peak, mode, threshold, line shape or correlation. This is an ontological statement, not a denial of the operational language used by experiments. The public object may be called an excess, a line-shape component, a resonance, a threshold feature or a correlation; the MMA-DMF reading asks whether it is a resolved response mode of an active medium rather than a new fundamental entity.

$$\text{stress/injection} \longrightarrow \text{structured response} \longrightarrow \text{resolved experimental mode.} \quad (32)$$

For the present paper the claim-bearing instance of Eq. (32) is the toponium-like threshold branch. The wider examples serve only as ontological continuity with the broader MMA-DMF program: high-energy astrophysical peaks, spectral-polarimetric features, medium-response signatures, and collider thresholds are not automatically independent fundamental entities. They are candidates for structured response modes whose interpretation depends on the relevant equation gate, source ledger and no-retuning status.

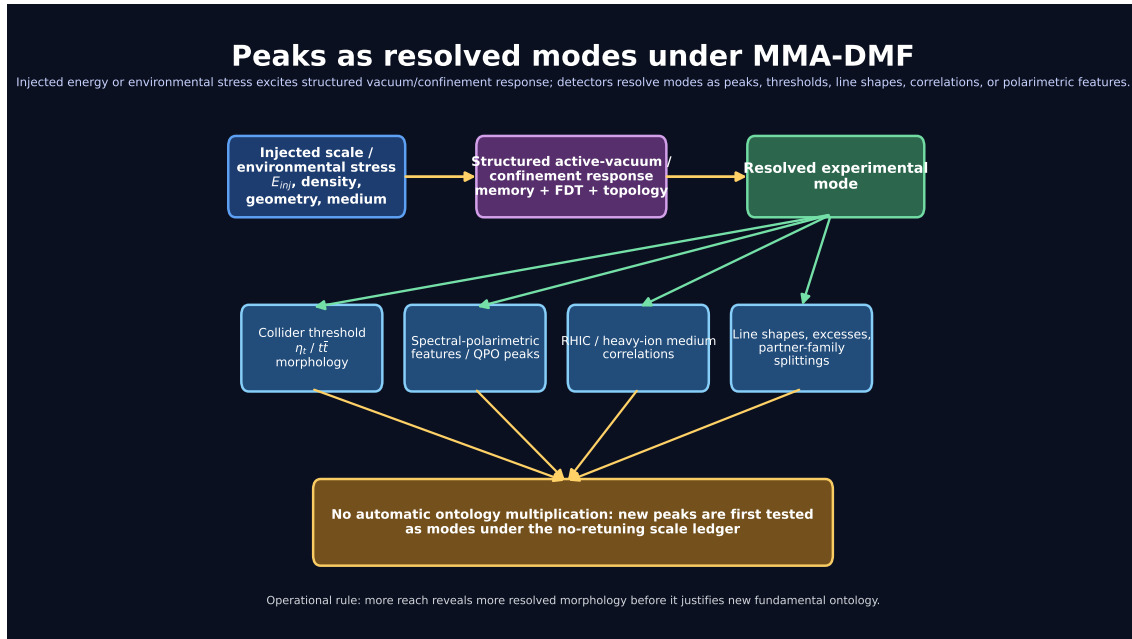


Figure 10: Peaks as resolved modes. MMA-DMF reads experimental peaks, thresholds, line shapes and correlations as detector-resolved manifestations of structured vacuum/confinement response under Eq. (32). The present paper uses this ontology only for the heavy-composite collider branch.

10 Energy-Evolution Prediction

The forward-looking statement is not the weak tautology that more data find more things. The MMA-DMF prediction is structured: increasing reach should populate common partner families, threshold refinements, spin-parity splittings, heavy-flavor multiplets, and rare channels that can be placed in the same ledger without adding independent ad hoc parameters. This follows from Eqs. (22), (18), (20), and (21). It does not require every newly resolved threshold or state to be fundamental. It predicts more resolved composite spectroscopy with a common classification rule.

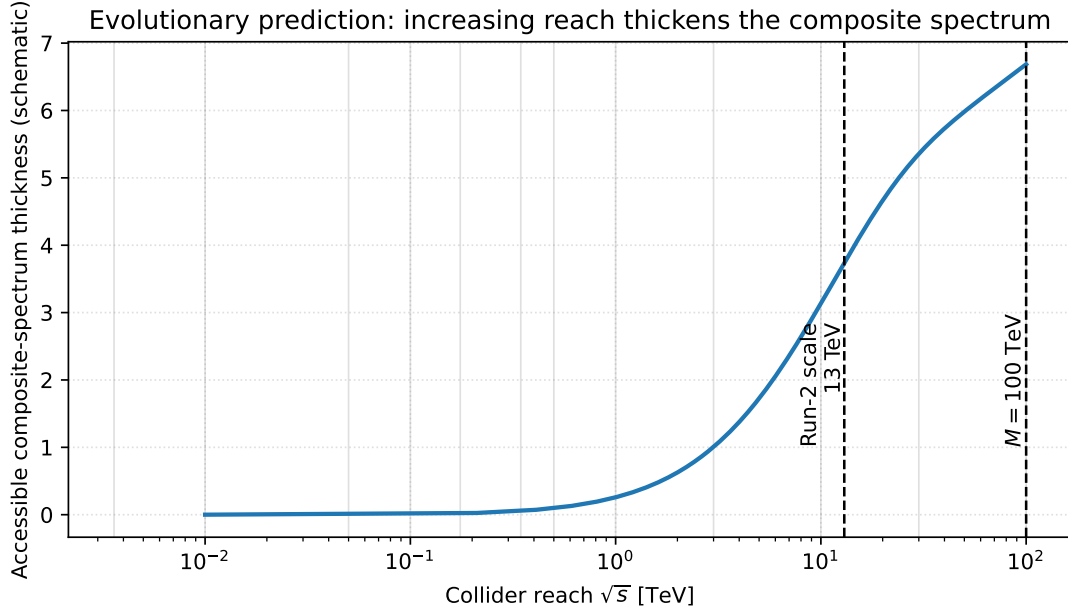


Figure 11: Evolutionary collider prediction. Increasing energy and luminosity should reveal more accessible composite states, threshold structures, heavy-flavor partners and rare channels below the one-scale closure. The curve is schematic: it states the predicted direction of accessible spectrum growth, not a per-state mass prediction.

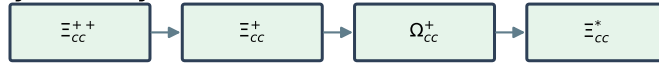
Structured collider-evolution prediction

Prediction: new public entries should thicken partner families and threshold refinements without independent ad hoc parameters per feature.

$t\bar{t}$ threshold family



ccq baryon family



higher reach

Pattern prediction, not a claim that every future structure is a new fundamental degree of freedom.

Figure 12: Structured version of the collider-evolution prediction. The figure emphasizes partner-family thickening and threshold refinement under Eqs. (22), (18), (20), and (21).

This prediction can be tested in several ways:

- additional $t\bar{t}$ threshold refinements or angular/parity separation at higher luminosity;
- improved η_t versus χ_t separation in public likelihood scans;
- more Ξ_{cc}^{++} and Ξ_{cc}^{+} channels and lifetime refinements;
- a final technical observation or exclusion path for Ω_{cc}^{+} ;

- excited ccq baryons and spin-parity assignments;
- additional heavy-flavor partner patterns that enter Eq. (2) without per-state retuning.

A failure mode is also clear. If increasing reach reveals only isolated features that require independent ad hoc parameters with no common mass/width/production/spin-parity ledger, the one-scale composite-spectroscopy reading weakens.

11 Claim-Bearing Test Ledger

Table 7: Compact claim-bearing ledger. The equation-link column points to the local gates actually used for each branch.

Branch	Public input	Equation links	Status
Toponium-like threshold	CMS TOP and HEPData η_t cross section and parity scan	Eqs. (1), (7), (8), (24), (21)	PASS_PUBLIC
Heavy scalar control	CMS HIG search with-/without η_t	Eqs. (26), (14)	PASS_GUARD
Ξ_{cc}^{++}	LHCb observation, PDG listing, lifetime/decay papers	Eqs. (2), (9), (27), (19)	PASS_PUBLIC
Ξ_{cc}^+ Ω_{cc}^+	LHCb Run-3 observation theory/lattice benchmarks	Eqs. (27), (28), (19) Eqs. (18), (20)	PASS_PUBLIC BENCHMARK
Official-data integrity layer	in- ZIP integrity manifest	Eq. (14); Fig. 2	PASS_INTEGRITY
Energy evolution	rising $\sqrt{s}$ and $\mathcal{L}$	Eqs. (22), (18), (21), (13)	PREDICTION
Detector-level reanalysis	none claimed	Eq. (14)	NOT_CLAIMED

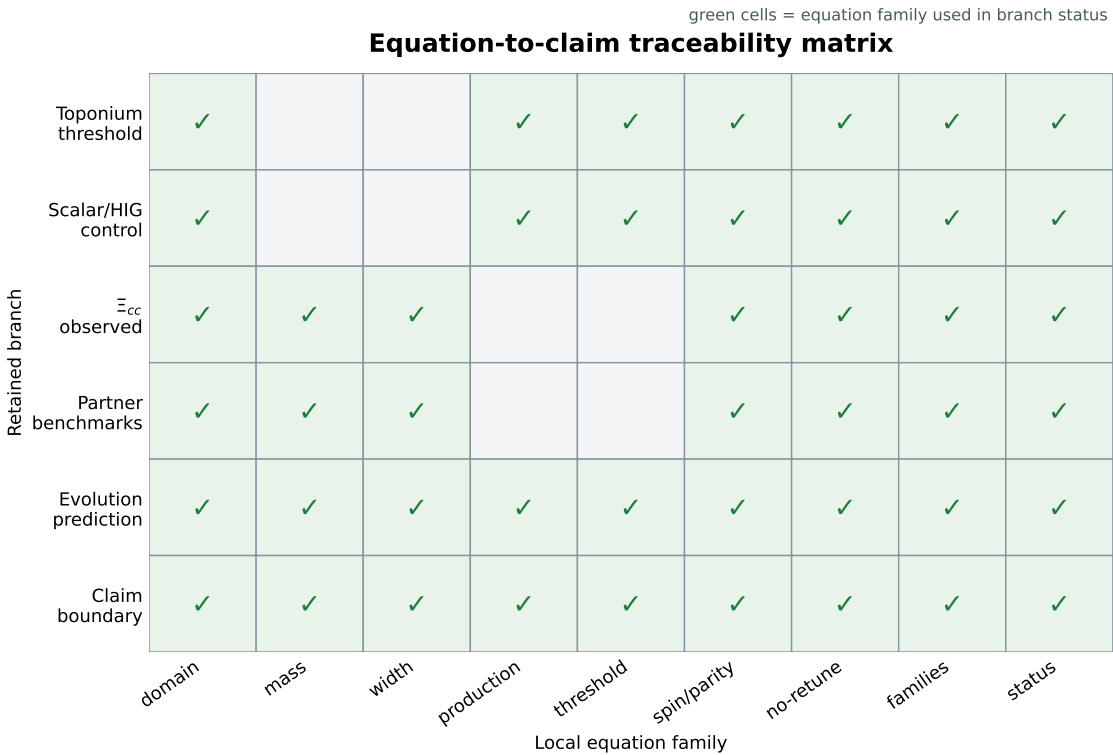


Figure 13: Equation-to-claim traceability matrix. The figure makes explicit that the retained branches are not floating interpretive statements; each passes through local equation gates and boundary status rules.

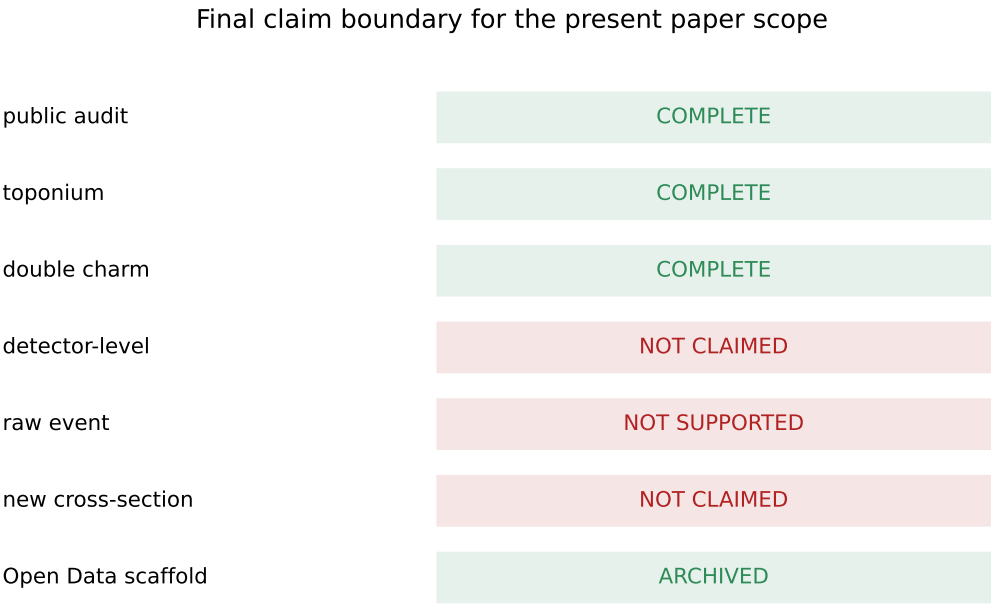


Figure 14: Final claim boundary. The official-data spectroscopy paper is complete for its declared scope while detector-level reanalysis, raw-event reprocessing and new cross-section extraction remain outside the manuscript.

12 Collider-Design Implication and Vacuum/Mass Context

If Eq. (17) is the retained discovery claim, the ideal experimental design is not simply maximum inclusive pp energy. The relevant design principle is controlled injected-energy scanning:

$$\frac{d\mathcal{M}^{\text{thr}}}{dE_{\text{inj}}} \quad \text{measured across the threshold,} \quad (33)$$

where $\mathcal{M}^{\text{thr}}$ denotes line-shape morphology, width response, parity separation and partner-family thickening. This is why top-threshold studies at future lepton colliders are relevant: a controlled e^+e^- threshold scan can target m_t , Γ_t and threshold-shape information much more directly than inclusive high-energy pp production [25]. This does not imply that LHC or CERN-style machines are wrong. It says that the decisive MMA-DMF scale-law test is a threshold-scan or medium-diagnostic experiment, not brute-force energy alone.

A RHIC-like logic enters as a complementary analogy: controlled collisions and medium diagnostics expose the response of QCD matter, vacuum correlations and hadronization. STAR reports evidence for spin correlation in $\Lambda\bar{\Lambda}$ pairs linked to QCD-vacuum quark-antiquark pair structure and confinement dynamics [26]. Heavy-ion jet-quenching and medium-response reviews likewise treat RHIC and LHC heavy-ion programs as probes of the quark-gluon plasma and medium transport, not merely as particle-production machines [27]. The MMA-DMF lesson is therefore precise: for composite spectroscopy, the machine of interest is the one that can vary the injected scale or characterize the medium well enough to map the response.

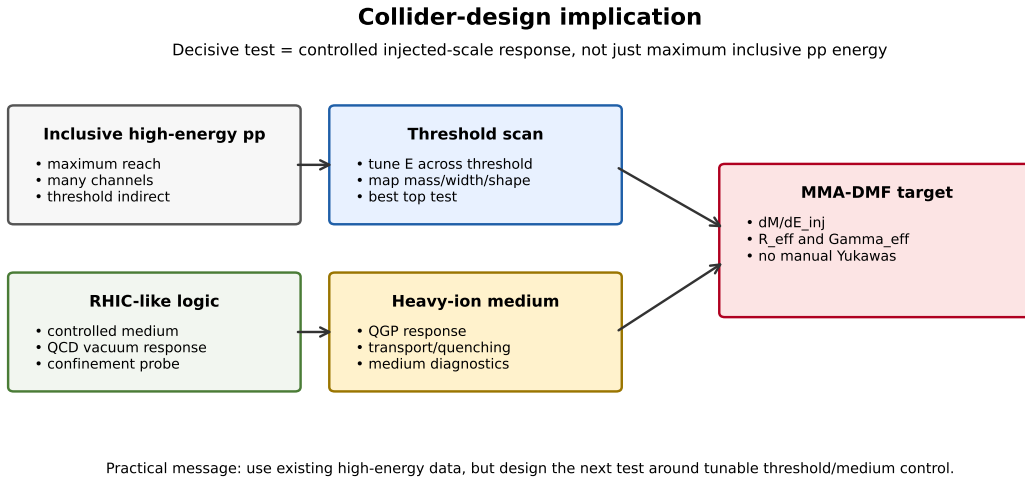
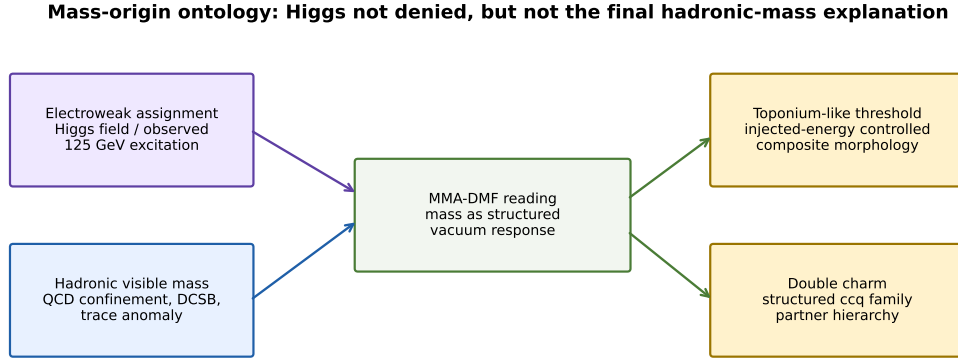


Figure 15: Collider-design implication of the scale-law claim. The paper does not argue that high-energy inclusive pp colliders are invalid. It argues that the decisive MMA-DMF test is controlled injected-energy or medium-response mapping: threshold scans for top-like morphology and RHIC-like tunability for vacuum/medium response.

Higgs ontology. The observed Higgs boson is not denied operationally. MMA-DMF denies its status as the final ontological origin of mass. It is interpreted as an effective breathing excitation of the active-vacuum order structure, while visible hadronic mass and composite-threshold morphology arise primarily from structured vacuum/confinement dynamics. This is not a rejection of the 125 GeV resonance; it is a demotion of manual Yukawa assignment from final explanation to operational bookkeeping.

The mass-origin context is similarly bounded. The observed Higgs boson is not denied. It remains the operational electroweak excitation observed at about 125 GeV. What is demoted is the idea that manual Yukawa assignment is the final ontological explanation of visible hadronic mass. Modern QCD already treats most visible mass as emergent from strong-interaction dynamics, dynamical chiral symmetry breaking, trace-anomaly structure and binding effects rather than from the sum of light current-quark masses [28, 29]. In MMA-DMF language, the Higgs is read as an effective breathing excitation of a deeper vacuum order parameter; hadronic mass and threshold morphology are structured vacuum/confinement responses.



The paper uses QCD/RHIC mass-emergence context to motivate a vacuum-response reading of composite spectroscopy.

Figure 16: Vacuum/mass context. The Higgs boson is not denied as an operational resonance. The MMA-DMF reinterpretation is ontological: visible hadronic mass and composite-threshold morphology are treated as structured vacuum/confinement response rather than as a list of manual Yukawa assignments.

This collider-design implication is one of the falsifiable upgrades of the paper. If future threshold scans or medium-controlled runs reveal additional structures but those structures require a fresh independent parameter at every step, the no-retuning scale-law reading fails. If instead the new structures thicken the same partner families, line-shape refinements and spin-parity splittings without new manual knobs, the scale-law claim is strengthened.

13 Discussion

The strongest part of the paper is not a large new numerical fit. It is a disciplined reclassification of a collider pattern. Increasing energy does not automatically create a separate ontology at every new feature. It opens phase space and detector reach. That reach thickens spectroscopy.

The toponium-like branch is a clean example because the public CMS interpretation already points toward quasi-bound $t\bar{t}$ threshold structure. The MMA-DMF contribution is to place that structure inside a one-scale closure and protect the interpretation against both extremes: denying the public threshold result, or overclaiming it as a new fundamental object.

The double-charm branch is complementary. The Ξ_{cc}^{++} and Ξ_{cc}^{+} entries are ordinary hadron-spectroscopy objects in public experimental language, but they are also exactly the kind of heavy-flavor composite states expected to become better resolved as collider data grow. Their role in this paper is not to prove a new topological mass calculation. It is to show that observed qqq heavy-flavor states fit the same composite gate as the $t\bar{t}$ threshold branch.

The Ω_{cc}^+ entry is intentionally bounded. Theory and lattice values make it useful as a benchmark, but unless a final technical observation paper is used, it is not written as an observed validation. This is the same claim discipline used elsewhere in the paper.

14 Limitations and Scope Boundaries

The paper has four main limitations.

1. It does not compute state-specific topological binding energies from first principles for every state. Residuals remain symbolic unless locked.
2. It does not perform detector-level analysis of CMS or LHCb events.
3. It does not claim raw-event reprocessing or new cross-section extraction.
4. It treats Standard-Model particle names as operational spectroscopy labels rather than fundamental ontology inputs.

These limitations are not hidden. They are part of the retained result. The intended scientific claim is a claim-bounded official-data composite-spectroscopy audit.

15 Data Availability

All public data used for claim-bearing numerical values are from CMS, LHCb, PDG, HEPData, and arXiv sources cited in the bibliography. The Overleaf package includes compact CSV excerpts derived from the CMS TOP HEPData record for the η_t cross section and parity scan, a supplementary source manifest, `ZIP_INTEGRITY_SHA256_MANIFEST.csv`, a public-file SHA256 ledger, and a `public_sources/` directory containing the public external files used by the paper. Large public CMS Open Data NANOAOB scaffolds and ZIP shards are not included in the Overleaf package because of size; they are retained as an external acquisition scaffold. The verified external NANOAOB package contains 20 ROOT/archive entries and 33.004 GB in four ZIP shards; all four archive parts passed integrity testing and have recorded SHA256 checksums.

For reproducibility, the minimal files to send with or deposit beside the preprint are: (i) this main PDF; (ii) the supplementary manifest PDF; (iii) the Overleaf source ZIP, excluding all internal equation PDFs and all internal 268-series archives; (iv) `public_sources/PUBLIC_FILE_LEDGER.csv`; (v) `supplement/ZIP_INTEGRITY_SHA256_MANIFEST.csv`; (vi) the four external public NANOAOB ZIP shards or their persistent Drive/Zenodo records; (vii) HEPData ins2905701 and ins2942928 packages; (viii) PDG/CMS/LHCb public PDFs used for numerical values; and (ix) the public-data acquisition Colab notebooks, clearly marked as acquisition/reproducibility scripts rather than detector-level analysis notebooks. This package is sufficient to reproduce the public-data provenance and the no-retuning ledger tables; it is not a claim to reproduce CMS detector-level selections.

16 Code Availability

The Overleaf package includes the LaTeX source, figures, compact CSV data excerpts used to generate selected figures, supplementary manifest tables, the public-file ledger, the ZIP-integrity manifest and Colab notebooks used for the public Open Data acquisition scaffold. Those notebooks pull and archive public files; they are not a detector-level analysis pipeline and are not used to claim a new cross-section extraction. The figures are generated from public numerical values, manifest summaries and schematic equations stated in this manuscript.

17 AI-Use Statement

AI-assisted drafting, formatting, LaTeX packaging, and figure-generation support were used under the author's direction. The author remains responsible for the final text, claims, equations, references, and scientific interpretation.

18 Conflicts of Interest

The author declares no institutional endorsement by CMS, LHCb, CERN, HEPData, PDG, or any experimental collaboration. Public collaboration data and publications are cited as public sources only.

19 Conclusions

This preprint reports a discovery-style organizing-principle result: a no-retuning injected-scale law for heavy composite spectroscopy. The CMS toponium-like threshold branch and the LHCb/PDG doubly-charmed partner-family branch are retained in one MMA-DMF composite ledger without manual Yukawa assignment or state-by-state parameter steering. The result does not claim a new experimental particle or collaboration endorsement; it claims that the public heavy-composite spectrum is more naturally read as scale-controlled vacuum/confinement morphology than as an expanding catalogue of manually assigned fundamental entities.

The nontrivial result is not that the retained states lie below 100 TeV. The nontrivial result is that threshold morphology, spin-parity structure, heavy-flavor partner family, public status and provenance can be locked into a common no-retuning ledger. The toponium-like branch is the main scale-law branch: the injected scale resolves a $t\bar{t}$ threshold response mode whose public line shape and parity morphology are retained without assigning a manual Yukawa knob. The double-charm branch is the control: ccu , ccd and ccs entries test whether the heavy-flavor partner hierarchy can be read as a family structure rather than isolated mass fits.

The Higgs boson is accepted operationally as an observed electroweak resonance. MMA-DMF rejects only the stronger ontological claim that manual Yukawa assignment is the final origin of mass. Visible hadronic mass and composite-threshold morphology are instead treated as structured vacuum/confinement response. This also changes the future experimental emphasis. If threshold morphology is controlled by injected scale, the decisive tests are controlled threshold scans and controlled medium diagnostics, not only higher-energy inclusive pp reach.

The public-data provenance layer is retained: the source chain uses CMS/LHCb/PDG/HEP-Data products and a four-part external NANOAOB scaffold with verified ZIP integrity, SHA256 hashes and 33.004 GB of archived public data. That provenance supports reproducibility and auditability. It does not imply collaboration endorsement, raw-event reprocessing, detector-level rediscovery or a new measured cross section.

The final retained status is

DISCOVERY_STYLE_NO_RETUNING_INJECTED_SCALE_LAW_WITHIN_DECLARED_OFFICIAL_DATA_SCOPE.

A Equation-to-Test Traceability

Table 8: Compact equation-to-test traceability sheet.

Equation	Role	Test or paper use
Eq. (1)	collider composite domain	proves all considered states are below $M = 100$ TeV
Eq. (2)	composite mass map	maps η_t , Ξ_{cc} and Ω_{cc} into a common mass gate
Eq. (3)	topological residual	blocks manual mass fitting by marking unlocked residuals symbolic
Eq. (4)	width and branching map	organizes lifetime/decay information where public
Eq. (5)	channel transition	marks state-specific transitions symbolic unless locked
Eq. (6)	public production ledger	connects cross-section observables to production language
Eq. (7)	threshold line shape	natural location of η_t threshold interpretation
Eq. (8)	$q\bar{q}$ spin-parity	toponium-like 0^{-+} branch
Eq. (9)	qqq spin-parity	doubly-charmed baryon classification
Eq. (10)	scale-ratio guard	documents distance from the 100 TeV closure scale
Eq. (13)	no-retuning rule	blocks per-state manual adjustment
Eq. (14)	status rule	assigns retained, benchmark or out-of-scope status
Eq. (18)	partner-family rule	organizes ccq states as a family, not isolated points
Eq. (19)	splitting hierarchy	states the isospin and strange-partner comparison
Eq. (21)	toponium threshold family	defines the $t\bar{t}$ threshold family and control directions
Eq. (22)	collider evolution	predicts structured accessible-spectrum thickening with reach
Eq. (29)	ccu/ccd splitting	hierarchy diagnostic for observed double-charm isospin pair
Eq. (30)	ccs strange-partner split	benchmark-only hierarchy diagnostic for Ω_{cc}^+
Eq. (32)	peaks-as-modes ontology	maps excesses, thresholds and line shapes to structured response modes
Eq. (33)	controlled injected-scale scan	future threshold-scan / medium-diagnostic falsification logic

B Extended Toponium Interpretation

The toponium branch is deliberately written as a threshold branch. Its strongest public input is not a table of individual raw events but a collaboration-level result, HEPData tables, and a public likelihood scan. The MMA-DMF question is therefore not whether an independent

analysis can reproduce every CMS nuisance parameter. The question is whether the public observable can be organized under a composite threshold equation. The answer is yes.

A false positive interpretation would be to say that the cross-section number in Eq. (24) has been derived by MMA-DMF. It has not. It is an external public observable. A second false positive would be to say that a new fundamental boson has been found. The paper blocks that via the CMS HIG control and Eq. (26). The retained interpretation is narrower and stronger: public data resolve a high-mass threshold structure that behaves like composite spectroscopy.

The scale ratio in Eq. (11) is central. Even the toponium-like threshold scale is a few parts in 10^{-3} of the closure scale. It therefore lies safely inside the composite regime. If future higher-luminosity or higher-energy datasets refine the angular correlations and line shape, the branch becomes more discriminating without requiring a new ontology.

C Extended Double-Charm Interpretation

The double-charm sector shows the same logic in a slower, baryonic spectroscopy regime. The observed Ξ_{cc}^{++} and Ξ_{cc}^+ states have masses near 3.62 GeV and enter the scale ratio at a few parts in 10^{-5} . Their discovery and channel refinement are a public example of spectroscopy thickening with experimental reach.

The *ccu* and *ccd* entries are not treated as separate fundamental principles. They are isospin partners in public hadron spectroscopy. Inside the MMA-DMF gate, they share the same form Eq. (27); differences arise from the light quark label, binding, spin and experimental/lifetime uncertainties. The *ccs* entry is included only as a benchmark because Ω_{cc}^+ has strong theoretical and lattice motivation but is not promoted to observed status by this paper.

D Public-Data Manifest Logic

The compact supplementary manifest lists the public files and records used by the paper. The Overleaf bundle now includes public source files, a public-file SHA256 ledger, the uploaded ZIP integrity manifest, and compact official-data audit tables. It intentionally does not include the large external ROOT ZIP shards or internal equation files. The main manuscript is self-contained: every equation needed to interpret a claim appears in Section 4. Public files are used for measured values and source provenance; they are not hidden dependencies.

E Reviewer-Facing Claim Checklist

A reviewer can check the paper using the following yes/no checklist.

1. Does the paper claim a detector-level reanalysis? No.
2. Does it claim raw-event reprocessing? No.
3. Does it claim a new cross-section extraction? No.
4. Does it cite the public CMS/LHCb/PDG/HEPData sources used for numerical values? Yes.
5. Does it define all MMA-DMF equations used locally? Yes.
6. Does it separate observed states from benchmarks? Yes.
7. Does it state an evolutionary prediction in falsifiable but bounded language? Yes.
8. Does it avoid institutional endorsement claims? Yes.

F ZIP Integrity Manifest Details

The uploaded `ZIP_INTEGRITY_SHA256_MANIFEST.csv` gives the reproducibility layer for the public-data pull used in the audit. It contains four rows, one per ZIP shard, with size, entry count, archive-integrity status, bad-file status and SHA256 hash. The full SHA256 strings are retained in the supplementary CSV included with the Overleaf package. To avoid an unreadable main-PDF table, the manuscript prints shortened hashes only.

Table 9: ZIP integrity manifest summary. Full SHA256 strings are included in `supplement/ZIP_INTEGRITY_SHA256_MANIFEST.csv`.

Archive	size GB	entries	ZIP OK	Short SHA256
part_0001.zip	9.191	6	True	58feb847...698575
part_0002.zip	9.778	7	True	38f50185...7645f3
part_0003.zip	9.640	5	True	c5a18ee9...d476a
part_0004.zip	4.395	2	True	569592bf...fc9df

The scientific boundary is explicit: this proves reproducibility and integrity of the local public-data pull, not collaboration endorsement and not a new detector-level measurement.

G Final Scope Freeze

The final manuscript status is:

$$\text{STATUS} = \text{DISCOVERY_STYLE_NO_RETUNING_INJECTED_SCALE_LAW_WITHIN_DECLARED_OFFICIAL_DATA_SCOPE.} \quad (34)$$

The scope is frozen at official-data spectroscopy. Attempts to upgrade the paper into a detector-level reanalysis, raw-event reprocessing, or new cross-section measurement require a separate analysis package and are not part of this manuscript.

H Expanded Public-Observable Tables

Table 10: Toponium official-data input table. Values are copied from the public CMS and HEPData layer; they are not refitted in this manuscript. The equation-link column prevents the table from becoming a loose narrative source list.

Observable	Public value or role	Equation links	Use in this paper
Collision system	$pp, \sqrt{s} = 13 \text{ TeV}$	Eqs. (16), (7)	public injected-scale environment
Luminosity	138 fb^{-1}	Eq. (6)	public CMS Run-2 exposure / production context
Threshold component	color-singlet pseudoscalar $^1S_0^{[1]} t\bar{t}$ quasi-bound state	Eqs. (8), (7), (21)	retained $q\bar{q}$ threshold branch
Measured excess cross section	$8.8^{+1.2}_{-1.4} \text{ pb}$	Eq. (24)	public observable, not a new extraction
Parity scan	$(\sigma_{\eta_t}, \sigma_{\chi_t}, 2\Delta\text{NLL})$ table	Eqs. (7), (21), (26)	parity/morphology discriminator
Heavy scalar/pseudoscalar search	CMS-HIG-22-013 control	Eq. (26)	anti-overclaim guard against new elementary-boson ontology

Table 11: Public double-charm input table. Observed and benchmark states are separated, and every row points to the local equation family used to assign status.

State	Content	Public status	Equation links	Paper use
Ξ_{cc}^{++}	<i>ccu</i>	observed; PDG listed	Eqs. (9), (27), (28)	observed public <i>qqq</i> partner-family entry
Ξ_{cc}^+	<i>ccd</i>	observed by LHCb Run 3	Eqs. (27), (28), (29)	isospin-partner hierarchy diagnostic
Ω_{cc}^+	<i>ccs</i>	theory/lattice benchmark	Eqs. (18), (20), (30)	benchmark-only strange-partner branch
Excited states	<i>ccq</i> <i>ccq</i> with <i>L, S, J</i> excitations	future spectroscopy	spec- Eqs. (22), (18)	spectrum-thickening path

I Detailed Future Falsification and Upgrade Path

The paper is designed to be upgradeable without rewriting its core claim. The upgrade conditions are intentionally strict.

I.1 Toponium branch upgrade

The toponium branch would strengthen if future public analyses supply finer threshold scans, improved angular correlations, public covariance matrices, and public likelihood surfaces that continue to prefer a quasi-bound threshold contribution over smooth continuum-only backgrounds. The branch would weaken if the excess disappears under improved modeling, if the parity-sensitive observables cease to favor the pseudoscalar threshold component, or if a high-mass new-particle interpretation becomes decisively favored by a public likelihood comparison. The present paper does not need that future outcome; it only states the current retained public-observable branch.

I.2 Double-charm branch upgrade

The double-charm branch strengthens with more Ξ_{cc}^{++} and Ξ_{cc}^+ decay channels, better lifetime measurements, spin-parity assignments, and observation or exclusion of Ω_{cc}^+ . It weakens if the public spectrum becomes incompatible with a common *ccq* composite ledger or requires state-by-state manual shifts not expressible as binding and spin terms in Eq. (27). The cleanest upgrade would be a public table of *ccq* ground and excited states with masses, widths, lifetimes, channels and spin-parity assignments.

I.3 Energy-evolution prediction upgrade

The energy-evolution prediction strengthens if future higher-luminosity or higher-energy runs resolve additional threshold states and heavy-flavor partners in a structured pattern. It weakens if additional features remain isolated and require unrelated explanation for each new resonance. This is the collider form of the balloon analogy: more energy increases the accessible internal volume of the same system; it does not by itself create a new balloon at every increment.

J Single Claim-Boundary Register

The paper uses a single non-claim register. Not claimed: CMS/LHCb endorsement, raw-event reprocessing, new cross-section extraction, detector-level rediscovery, or discovery of a new experimental particle by this author. These exclusions do not weaken the scale-law claim; they specify its evidence class. The retained claim is the organizing no-retuning injected-scale law stated in Eq. (17) and frozen in the final status line.

K Long-Form Interpretation of the Final Claim Set

The final claim set should be read as a narrow spectroscopy result. The paper does not ask whether all of particle physics should be rewritten from collider observables alone. It asks whether two public heavy-sector signals - one $t\bar{t}$ threshold branch and one double-charm baryon branch - can be organized under one composite-spectroscopy closure without adding a new ad hoc degree of freedom for each observed feature.

The answer is affirmative because the two branches occupy complementary regions of the same argument. Toponium tests the high-mass threshold side: a near- $2m_t$ line-shape structure at hundreds of GeV. Double charm tests the heavy-flavor baryon side: ccq states at a few GeV. Both are far below 100 TeV and both are naturally composite in public experimental language. That is exactly the domain in which a one-scale composite closure should operate.

The paper is also deliberately conservative. It does not try to convert every public label into a direct MMA-DMF proof. The mass closure marks state-specific residuals symbolic unless the topological binding and spin terms are locked. The width equations mark channel transitions symbolic unless the transition operator is locked. The production equation accepts public cross sections but does not claim to reproduce detector-level selections. This conservative discipline is why the result is defensible.

L Publication-Level Reproducibility Notes

The Overleaf package contains the LaTeX source, figures, compact HEPData-derived CSV excerpts and compact manifest tables. It does not contain the large NANOAOB ZIP shards because those are tens of gigabytes. It also does not contain internal equation ledgers. The publication-level package is therefore suitable for arXiv, Zenodo, SSRN or ScienceOpen submission without accidentally bundling private/internal materials or unwieldy raw archives.

M Final Editorial Reading

The paper's headline should be understood as follows. "Toponium" names the public threshold-spectroscopy branch. "Doubly-charmed spectroscopy" names the public ccq partner-family branch. "One-scale hadronic closure" names the local equation spine that lets both branches be read through one no-retuning scale map. The title does not claim a new CMS/LHCb particle discovery, collaboration endorsement, or raw-data reanalysis. It claims a discovery-style organizing law: manual Yukawa/state-by-state retuning is rejected, while injected scale, spin-parity structure, public state labels and partner-family hierarchy control the retained composite-spectroscopy morphology.

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